ASTRID BEYER

3D PROGRAMMER | GRAPHICS & COMPUTATIONAL GEOMETRY

La Ciotat, France | +33 6 75 44 42 49 | astrid.beyer@orange.fr | teloru.dev | linkedin.com/in/astrid-be | github.com/Teloru

PROFILE

3D R&D engineer specializing in computational geometry, mesh processing and production 3D pipelines. Own a geometry product end-to-end, from ML segmentation to manufacturable output. C++/Python background with hands-on graphics experience in OpenGL, GLSL, WebGL/Three.js and game development.

CORE SKILLS

Languages: C++, Python, GLSL, JavaScript / TypeScript

3D & Graphics: Computational geometry, mesh processing, OpenGL, WebGL, Three.js, real-time rendering, shaders, rendering pipelines

Libraries: PyMesh, libigl, Trimesh, Shapely, TTK, SFML

Systems: Docker, CI/CD, performance optimization, debugging & profiling, clean production code

EXPERIENCE

H3D | 3D R&D Software Engineer, Dental Team

Copenhagen, Denmark (remote) | Dec 2024 - Present

- Own the Models & Retainers product line end-to-end, designing and shipping automated 3D mesh-processing pipelines that turn raw intraoral scans into production-ready dental models and retainers.
- Built reconstruction logic on top of ML segmentation, converting segmented scans into watertight, manufacturable geometry using Python-based geometry tooling.
- Led the Hugin optimization project, reducing ML pipeline runtime by 5-10% and cutting more than one hour from large CI/CD test-set jobs; coordinate releases across AI, full-stack and product teams.

Ubisoft Montpellier | Mentee, Global Mentoring Program

Castelnau-le-Lez, France | Oct 2024 - Mar 2025

- Developed a Tower Defense game in C++ / SFML under a Lead Gameplay Programmer, using a modular, performance-oriented ECS architecture.
- Implemented an isometric rendering system without an engine using draw order and spatial sorting to simulate depth; documented technical design decisions in a YouTube walkthrough.

Dassault Systèmes | Software Engineer Intern, Additive Manufacturing

Aix-en-Provence, France | Apr 2024 - Sep 2024

- Built a configurable G-code post-processing pipeline in 3DEXPERIENCE for multiple printers, translating geometric toolpaths into machine-specific instructions.
- Reverse-engineered APT/G-code variants, tuned toolpath parameters to mitigate fabrication defects, and built a flexible UI for machine-dependent configuration.

LIS Laboratory | Computer Graphics Researcher, G-Mod Team

Marseille, France | Apr 2023 - Jun 2023

- Optimized TTK-based topological extraction on 3D meshes, added STL/PLY support and topological filters, cutting mesh-processing time by 20% and reducing C++ compatibility errors by 30%.

EDUCATION

M.Sc. Computer Graphics & Applied Geometry | Aix-Marseille University | 2024 | With Merit, ranked #2 | GPA 3.7

B.Sc. Computer Science | Aix-Marseille University | 2022 | Major: GPU Programming

Associate Degree, Computer Science | Aix-Marseille University | 2020 | Software development, UNIX, Qt